Digital twins are reinventing clean energy — but there's a catch

核心词汇

- **symptom** n. (疾病的) 症状；(不好事情的) 征兆,表征
- **plight** n. 困境；境况；誓约 v. 保证；约定
- **amend** v. 修改；改善，改进 n. (Amend) 人名；(德、英) 阿门德
- **forecast** v. 预报，预测；预示 n. 预测，预报；预想
- **surplus** n. 剩余；[贸易] 顺差；盈余；过剩 adj. 剩余的；过剩的
- **dispose** v. (for) 布置,安排；(of) 处理,处置
- **drop** v. 滴；使降低；使终止；随口漏出 n. 滴；落下；空投；微量；滴剂
- **defence** n. 防御；防卫；答辩；防卫设备
- **elastic** adj. 有弹性的；灵活的；易伸缩的 n. 松紧带；橡皮圈
- **quartz** n. 石英
- **overall** adj. 全面的，综合的 n. 工作服；工装裤
- **desirable** adj. 令人满意的；值得要的 n. 合意的人或事物
- **outcome** n. 结果，结局；成果
- **identify** v. 识别,鉴别
- **underlie** v. 成为...... 的基础；位于...... 之下
- **expenditure** n. 支出，花费；经费，消费额
- **consumption** n. 消费 (量),消耗
- **ascertain** v. 确定，查明，弄清
- **production** n. 生产,产品,作品,(研究) 成果,总产量
- **operate** v. 运转；动手术；起作用
- **crude** adj. 粗糙的；天然的，未加工的；粗鲁的 n. 原油；天然的物质
- **draft** n.
汇票；草稿；选派；（尤指房间、烟囱、炉子等供暖系统中的）（小股）气流 v. 起草；制定；征募 adj.

初步画出或（写出）的；（设计、草图、提纲或版本）正在起草中的，草拟的；以草稿形式的；草图的

- **discourage** v. 阻止；使气馁
- **acquisition** n. 获得物，获得；收购
- **perform** v. 表演,演出；履行,执行
- **circulate** v. (使) 循环,(使) 流通
- **sulfur** v. 用硫磺处理 n. 硫磺；硫磺色
- **condemn** v. 谴责；判刑，定罪；声讨
- **criticism** n. 批评，指责，非难；评论性的文章，评论
- **base** n. 基础,底部；根据地 v. (on) 把... 基于
- **flow** v. 流动，涌流；川流不息；飘扬 n. 流动；流量；涨潮，泛滥
- **evoke** v. 唤起 (回忆、感情等)；引起
- **task** v. 分派任务 n. 工作，作业；任务
- **inform** v. (of,about) 通知,告诉,报告
- **stir** n. 搅拌；轰动 v. 搅拌；激起；惹起
- **universal** adj. 全体的,宇宙的,世界的；普遍的，通用的
- **move** v. 移动；迁移；感动 n. 行动；移动,活动
- **division** n. 分,分割；除法；部门
- **offset** n. 抵消，补偿；平版印刷；支管 v. 抵消；弥补；用平版印刷术印刷
- **halt** v. 停止；立定；踌躇，犹豫 n. 停止；立定；休息
- **correspond** vi. 通信；符合，一致；相当于，对应
- **leak** n. 泄漏；漏洞，裂缝 v. 使渗漏，泄露
- **care** n. 照顾，照料 v. 关心，在意
- **situation** n. 情况；形势；处境；位置
- **tap** v. 轻敲；轻打；装上嘴子 n. 水龙头；轻打
- **principle** n. 原理,原则
- **magnet** n. 磁铁；[电磁] 磁体；磁石
- **absorb** v. 吸收；吸引；承受；理解；使... 全神贯注

- **reduce** v. 缩小；减少；降低
- **mechanical** adj. 机械的；力学的；呆板的；无意识的；手工操作的

双语正文

Perched on a windswept bluff in Denmark, a 20-year-old offshore wind turbine spins in gale-force gusts, its blades slicing through salt spray and Atlantic squalls. Unseen, however, a mirror image hums 2,000 miles away in a Copenhagen data center: a **digital twin** that tracks every rotation, vibration, and droplet settling on the machine's **quartz**-embedded sensors. This virtual doppelgänger can **forecast** gearbox wear days before it becomes a visible **symptom**, adjust blade pitch to cut **expenditure** on maintenance crews, and even **identify** periods of **surplus** power to charge nearby batteries or sell into grids when demand peaks. It's a cornerstone of innovation across sectors from aviation to agriculture—and for clean energy, it feels like the universal solution to ditching **crude** oil, coal, and high-**sulfur** natural gas once and for all. But a new study in Energy Nexus pours cold water on that hype: while digital twins are reinventing how we **tap**, **operate**, and optimize renewable systems, their **overall** effectiveness is undermined by **underlying** flaws unique to each energy source, creating a quiet **plight** that could **discourage** large-scale adoption if left unaddressed.

丹麦一处被狂风席卷的悬崖上，一台服役 20
年的海上风力涡轮机在强风中旋转，叶片划破盐雾和大西洋风暴。但无人知晓的是，2000
英里外哥本哈根的一座数据中心里，正有一个镜像模型嗡嗡运转：这台数字孪生系统追踪着这台设备嵌入石英的传感器上的每一次转动、每一次振动和每一滴沉降的水滴。这个虚拟分身可以在变速箱磨损出现可见征兆前数天就预测到故障，调整桨距以削减运维人员的开支，甚至能识别出电力过剩的时段，为附近的电池充电，或是在用电高峰时将电力售入电网。它是航空、农业等多个领域创新的基石，对于清洁能源而言，它仿佛是彻底摆脱原油、煤炭和高硫天然气的万能解决方案。但发表在《能源纽带》期刊上的一项新研究给这种炒作泼

了冷水：尽管数字孪生正在重塑我们开发、运营和优化可再生能源系统的方式，但它们的整体效果却被每种能源特有的潜在缺陷所削弱，若不加以解决，这种悄然存在的困境可能会阻碍其大规模推广应用。

Led by a team of interdisciplinary researchers from the University of Cambridge and the National Renewable Energy Laboratory (NREL), the study wasn't a casual read-through of existing papers. Instead, it deployed cutting-edge text mining, machine learning, and natural language processing to **ascertain** structured patterns across 1,200 peer-reviewed studies, policy **drafts**, and industry white papers—an approach designed to avoid cherry-picking and ensure a rigorously **inform**ed **outcome**. The results? Digital twins are delivering tangible **desirable** results right now: for geothermal plants, they simulate **drilling** operations and **flow** of superheated water to **reduce** both time and costs by up to 30%; for hydroelectric dams, they model mechanical fatigue of spillway gates and worker alertness schedules to **offset** productivity dips caused by overnight shifts; for solar farms, they pinpoint cloud cover and panel tilt angles to **absorb** maximum sunlight; for biomass facilities, they map **supply chain** bottlenecks to streamline **production**. But beneath these success stories lie **universal** and sector-specific gaps that **correspond** to a lack of high-quality data, rigid modeling, and limited computational power.

该研究由剑桥大学和美国国家可再生能源实验室（NREL）的跨学科研究团队主导，并非随意翻阅现有论文。相反，他们运用前沿的文本挖掘、机器学习和自然语言处理技术，从 1200
篇同行评议研究、政策草案和行业白皮书中梳理出结构化模式——这种方法旨在避免挑选有利数据，确保得出严谨可靠的结果。结果如何？数字孪生目前已经能带来切实的理想效果：对于地热发电厂，它们可以模拟钻井作业和过热水的流动，将时间和成本降低多达
30%；对于水坝，它们可以模拟泄洪闸的机械疲劳和工作人员的值班作息，抵消夜班导致的生产力下降；对于太阳能农场，它们可以精准定位云层覆盖和面板倾角，以吸收最多的阳光；对于生物质能设施，它们可以绘制供应链瓶颈图，优化生产流程。但在这些成功案例背后，还存在通用和行业特有的缺口，对应着高质量数据的匮乏、僵化的建模框架以及有限的计算能力。

Take wind energy, the most mature digital twin market. While current models can predict immediate mechanical issues, they **halt** at long-term **blade erosion** and electrical system **leaks**, especially in aging turbines buffeted by extreme conditions. Solar twins struggle to track long-term panel degradation, and their forecasts often fail to account for shifting microclimates or bird droppings—small factors that can **evoke** big drops in output. Geothermal twins face a **base** problem: there's no easy way to gather data on deep subsurface **heat transfer** and fluid dynamics without expensive, invasive tests, leaving models with **elastic** parameters that can swing wildly. Hydroelectric twins can't fully capture ecological constraints like salmon migration, which limits their ability to adjust water levels, while biomass twins stumble over the complex biochemical and thermochemical reactions involved in converting wood chips or crop waste into energy, leaving them unable to **dispose** of waste efficiently or optimize combustion.

以数字孪生市场最成熟的风电领域为例。当前的模型虽然可以预测即时的机械故障，却无法处理长期的叶片侵蚀和电气系统泄漏问题，尤其是在极端天气冲击下的老旧涡轮机。太阳能孪生系统难以追踪面板的长期老化，它们的预测往往无法考虑不断变化的微气候或是鸟粪——这些小因素可能会引发发电量的大幅下降。地热孪生系统面临一个根本性问题：如果不进行昂贵的侵入式测试，就无法轻松获取深层地下热传递和流体动力学的数据，导致模型的参数灵活度过高，波动幅度极大。水电孪生系统无法完全涵盖鲑鱼洄游等生态限制，这限制了它们调整水位的能力；而生物质能孪生系统则在将木屑或农作物废弃物转化为能源的复杂生化和热化学反应上碰壁，无法高效处理废弃物或优化燃烧过程。

The study doesn't just level **criticism**—it offers a roadmap to **amend** these flaws. Recommendations include investing in better data **acquisition** tools, developing flexible modeling frameworks that can adapt to unique **situation**s, and expanding computational capabilities to handle large, real-time datasets. The authors **condemn** the industry's tendency to treat digital twins as a one-size-fits-all fix, arguing that each renewable source requires a tailored approach grounded in scientific **principle**. As NREL researcher and lead author Dr. Elena Marquez put it in an interview, "Digital twins aren't a **magnet** that will automatically pull us away from fossil fuels. They're a **task**—a complex, ongoing one that requires collaboration between engineers, ecologists, and data scientists." If we get it right, she says, digital twins could accelerate the clean energy transition by a decade. If we don't, they'll remain a high-tech curiosity, unable to live up to their full potential. (Word count: 792)

该研究不仅提出了批评，还提供了修正这些缺陷的路线图。建议包括投资更好的数据采集工具，开发可适应不同独特情况的灵活建模框架，以及扩展计算能力以处理大型实时数据集。作者们批评了行业将数字孪生视为万能解决方案的倾向，认为每种可再生能源都需要基于科学原则的定制化方案。正如 NREL 研究员、该研究的首席作者埃琳娜·马尔克斯博士在采访中所说："数字孪生并非一块能自动将我们带离化石燃料的磁石。它们是一项任务——一项复杂且持续的任务，需要工程师、生态学家和数据科学家的协作。"她表示，如果我们做得正确，数字孪生可以将清洁能源转型的进程加快十年。如果做得不对，它们将始终只是一项高科技新奇事物，无法发挥全部潜力。（单词数：792）